

GIACOMINA (MINA) GRANDOLFO

PhD Candidate in Cognitive Neuroscience

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DOCTORAL RESEARCH

PhD candidate — Interoception, metacognition and lucid dreaming

Center for Research in Cognition & Neurosciences (CRCN), Université Libre de Bruxelles

2023 – present

Brussels, Belgium

- **Experiment 1** — Cardio-visual enfacement illusion ($n = 47$; 22 fLD / 25 nLD). Manuscript in preparation.
- **Experiment 2** — Heartbeat Discrimination Task ($n = 53$). Interoceptive metacognition via HMeta-d.
- **Experiment 3** — Rubber Hand Illusion ($n > 55$). Proprioceptive drift and body ownership in lucid vs. non-lucid dreamers.
- **Experiment 4** — Virtual Reality Rod-and-Frame task (Unity, HTC Vive). Bayesian visuo-vestibular integration (Alberts 2016) with adaptive Psi marginal (Prins 2013).

Supervisors

Axel Cleeremans — Doctoral supervisor (director)

Consciousness, Cognition & Computation Group, ULB

Theoretical framework on consciousness and global workspace.

Charline Urbain — Co-supervisor

Université Libre de Bruxelles

Neuroimaging specialist (fMRI & MEG).

Peter Simor — Supervisor

Department of Cognitive Psychology, Eötvös Loránd University, Budapest

Day-to-day collaborator, neuroscience of sleep.

Ongoing collaborations

Audrey Mazancieux - Scientific collaborator

Université Libre de Bruxelles / CRCN

2025 – 2026

- Co-author on the Heartbeat Discrimination Task manuscript (interoceptive metacognition).
- Joint literature review on metacognition and lucid dreaming (summer 2026).

François Foerster — Postdoctoral researcher

Consciousness, Cognition & Computation (CO3) group, ULB

2026

- Collaboration on the Virtual Reality Rod-and-Frame protocol (Experiment 4).

PUBLICATIONS & MANUSCRIPTS

Grandolfo, G., Urbain, C., Cleeremans, A., & Simor, P. (in prep). Lucid dreamers resist the cardio-visual enfacement illusion: interoceptive accuracy moderates multisensory self-other boundaries.

RESEARCH INTERESTS

Predictive processing Metacognition
Interoception Lucid dreaming
Multisensory integration Self-model
Body ownership Consciousness
Computational modelling

EDUCATION

PhD in Cognitive Neuroscience

Université Libre de Bruxelles – CRCN

2023 – present

Thesis: Interoception, metacognition and lucid dreaming. Supervisors: A. Cleeremans, C. Urbain, P. Simor.

Master's in Psychological Sciences — Research track in Neuropsychology

Université Libre de Bruxelles

2021 – 2023

Highest distinction (summa cum laude), 18/20. Erasmus exchange, Universidad de Sevilla.

Bachelor's in Psychology

Université Libre de Bruxelles

2018 – 2021

Computer Science (2 years)

IEPSCF Uccle, Brussels

2020 – 2022

Software development & networking — programming, databases, networks, systems administration.

Master's in Labour Sciences

Université Libre de Bruxelles

2014 – 2016

Great distinction (magna cum laude).

Bachelor's in Special Education

HEB Defré, Brussels

2011 – 2014

Great distinction (magna cum laude).

Grandolfo, G., Mazancieux, A., Urbain, C., Cleeremans, A., & Simor, P. (in prep). Enhanced interoceptive metacognition in frequent lucid dreamers: evidence from the heartbeat discrimination task.

Grandolfo, G. & Mazancieux, A. (in prep). Metacognition and lucid dreaming: a literature review.

PRESENTATIONS

Talks

2026 — *Lucid dreaming modulates the cardio-visual enfacement illusion.* ASSC29, Santiago, Chile (abstract accepted).

Posters

2026 — *Interoceptive precision and self-other boundaries in lucid dreamers.* Neurosciences of Dreaming, Lucca, Italy (accepted).

2026 — *Lucid dreaming, interoceptive precision and the enfacement illusion.* ESRS — European Sleep Research Society (abstract submitted).

2025 — *Lucid dreaming, interoception and multisensory integration: evidence from waking paradigms.* UNI General Meeting, ULB.

2024 — *Lucid dreaming and self-recognition: interoceptive signals modulate body ownership illusions.* ASSC27, Tokyo.

Other scientific activities

2024 — Co-organiser, *Cognitive Control & Performance Monitoring* symposium (3 days, Wim Gevers, ULB).

2025 — CuttingEEG workshop (attendee — MNE-Python, M/EEG source analysis).

2023 — NeuroCog symposium, Brussels (attendee).

EARLIER PROFESSIONAL EXPERIENCE

Managing Director & Technical Project Manager

AVS Elektro SPRL

📅 2014 – 2017 📍 Brussels, Belgium

Home automation, sound systems, AutoCAD technical drawings, team coordination, financial oversight.

Communications officer

Amaragea ASBL — youth service, Brabant Wallon

📅 2011 – 2014

Webmaster, content creation, project management.

Educator (*éducatrice*)

Centre Ferme Canada & Prison d'Everberg (minors)

📅 2009 – 2011

Psychological support in residential facilities; individual interventions.

TEACHING & SUPERVISION

Research intern supervisor

CRCN, ULB

📅 2024 – 2026

6 interns across multiple experiments; protocol training, data quality monitoring, evaluations.

MSc thesis supervision

ULB — *Lucid dreaming and metacognition*

📅 2025 – 2026

Student teaching assistant —

Neuropsychology

Université Libre de Bruxelles

📅 2021

Guidance sessions (2 h/week): reformulation, Wooclap quizzing, Q&A.

Student delegate — Bachelor's

Psychology

Université Libre de Bruxelles

📅 2020

TECHNICAL SKILLS

Programming

Python R C# Unity

Methods

Signal Detection Theory HMeta-d
Bayesian modelling Adaptive Psi
Mixed-effects models Bayes factors

Experimental platforms

HTC Vive BioSemi ECG/EEG OSF
PsychoPy

LANGUAGES

French
Italian
English
Spanish



REFERENCES

Available on request.